Association between the relative abundance of butyrate-producing and mucin-degrading taxa and Parkinson's disease

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Highlights

1. We explored the gut microbiome in a South African Parkinson's disease (PD) cohort.

2. Beta-diversity had a significant association with PD status (q = 0.008).

3. Butyrate-producing bacteria (*Faecalibacterium* and *Roseburia*) were reduced in PD cases.

4. An increase in mucin-degrading bacteria (*Akkermansia*) was found in PD cases.

5. These alterations might be associated with altered gut permeability and inflammation.

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Abstract

Background

Parkinson's disease (PD) is a neurodegenerative disorder characterised by motor and non-motor symptoms. Differences in the gut microbiome composition and diversity have been implicated in PD aetiology. This study aimed to further explore the interplay between the gut microbiome and PD in the South African context.

Methods

Gut microbial data (cases: n = 16; controls: n = 42) was generated using a 16S rRNA gene (V4) primer pair. The DADA2 pipeline was used for data preparation and taxa classification. Alpha-diversity was assessed using the Shannon and Simpson diversity metrics, whereas beta-diversity was assessed with the permutational multivariate analysis of variance (PERMANOVA) adonis in QIIME2. Differentially abundant taxa between cases and controls were evaluated using Analysis of Compositions of Microbiomes with Bias Correction (ANCOM-BC).

Results

Alpha-diversity was not associated with PD status, however, beta-diversity was associated with PD (q = 0.008). Furthermore, PD was associated with the depleted relative abundance of *Faecalibacterium* (q = 0.007), *Roseburia* (q = 0.030), *Dorea* (q = 0.015), and *Veillonella* (q < 0.001), and the enriched relative abundance of *Akkermansia* (q = 0.015) and *Victivallis* (q < 0.001).

Conclusion

Our study found gut microbiome alterations with a reduction in butyrate-producing bacteria (e.g. *Faecalibacterium* and *Roseburia*) and an increase in mucin-degrading bacteria (*Akkermansia*) in PD cases compared to controls. These alterations might be associated with heightened gut permeability and inflammation. Longitudinal studies can address the question of whether these microbiome differences are a risk factor for or are consequent to the development of PD.

Introduction

Parkinson's disease (PD) is a progressive, debilitating disorder characterised by a range of motor and non-motor symptoms (Diaz et al., 2022). The prevalence of PD is estimated between 0.3% and 1% among individuals over 60, with a male:female incidence ratio of 3:1 (Romano et al., 2021). In developing regions such as Africa, Asia, and Latin America, the prevalence of PD is estimated to be lower due to environmental differences, genetics, healthcare availability and lower life expectancy (Ben-Shlomo et al., 2024). Non-motor symptoms, such as gastrointestinal symptoms, typically present as a marker of preclinical autonomic

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dysfunction in PD, suggesting the enteric nervous system (ENS) may be compromised prior to the central nervous system (CNS) (Romano et al., 2021; Ryman et al., 2023). The trillions of microbes in the gut and their genetic material, known as the gut microbiome, facilitate bidirectional communication between the ENS and the brain, forming the microbiome-gut-brain (MGB) axis (Cryan et al., 2019).

There are currently two hypotheses on how gastrointestinal dysfunction influences the MGB axis to potentially contribute to the progression of PD (Ryman et al., 2023). As PD is caused by an abnormal accumulation and aggregation of alpha-synuclein (α-synuclein) in the CNS (Nishiwaki et al., 2020; Hirayama & Ohno, 2021), the first focuses on abnormal aggregation of α-synuclein transported from the gut via the vagus nerve, to the CNS, the lower brainstem, and the cerebral cortex (Nishiwaki et al., 2020; Ryman et al., 2023). Differences in the gut microbiome can cause elevated levels of lipopolysaccharides (LPS) that contribute to higher α-synuclein levels in the gut (Lei et al., 2021; Ryman et al., 2023), α-synuclein misfolding or abnormal aggregation of α-synuclein (Lei et al., 2021). Transportation of the abnormally aggregated α-synuclein to the brain (Nishiwaki et al., 2020; Ryman et al., 2023) could lead to the presence of eosinophilic Lewy bodies in the cytoplasm of dopaminergic neurons and also to mitochondrial dysfunction of dopaminergic neurons (Lei et al., 2021). Misfolded α-synuclein can also stimulate an inflammatory response in microglia, which can increase oxidative stress and intensify abnormal α-synuclein aggregation, thereby forming a positive feedback loop (Lei et al., 2021). The second hypothesis focuses on chronic intestinal pro-inflammatory processes as a key contributor to gastrointestinal dysfunction and disease progression of PD, discussed below (Ryman et al., 2023).

Both hypotheses emphasise that imbalances in gut microbiota and intestinal permeability are key factors in PD development (Takahashi et al., 2022; Li et al., 2023; Ryman et al., 2023). Thereby implicating gut microbial alterations in the pathogenesis of PD by contributing to α-synuclein changes (Ryman et al., 2023), increased gut permeability, and inflammation (Aho et al., 2021; Romano et al., 2021). Patients with PD exhibit increased gut permeability and inflammation (Aho et al., 2021; Romano et al., 2021), which are associated with lower levels of short-chain fatty acid (SCFA)(Aho et al., 2021) and high LPS levels produced by gut microbiota (Hirayama & Ohno, 2021; Romano et al., 2021). These conditions can trigger inflammatory responses (Lei et al., 2021; Ryman et al., 2023), increase blood-brain barrier (BBB) permeability (Ryman et al., 2023), and promote the bidirectional movement of α-synuclein between the gut and brain (Nishiwaki et al., 2020; Hirayama & Ohno, 2021).

Research indicates that gut microbial communities play a role in the development and severity of PD, with altered microbiomes and decreased bacterial diversity observed in both animal models and human patients (Aho et al., 2021; Kang et al., 2021). However, no single microbial profile has been identified that is associated with PD aetiology (Takahashi et al., 2022; Li et al., 2023; Ryman et al., 2023). Alterations in the gut microbiome include decreased relative abundance of butyrate-producing taxa (*Roseburia*,

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Faecalibacterium, Lachnospiraceae)(Nishiwaki et al., 2020; Hirayama & Ohno, 2021; Lei et al., 2021; Romano et al., 2021; Takahashi et al., 2022) and an increase in the relative abundance of taxa relating to mucin layer-degrading and LPS producing taxa, such as *Akkermansia* (Aho et al., 2021; Takahashi et al., 2022; Ryman et al., 2023), contributing to increased intestinal permeability, oxidative stress, and α-synuclein buildup in the ENS (Kang et al., 2021). Levels of SCFA are associated with microbial diversity, showing negative correlations with *Bifidobacterium* and positive correlations with *Bacteroides* in control groups compared to PD cases, highlighting the functional impact of gut microbiome differences on PD (Aho et al., 2021).

While environmental, genetic, and microbial factors contributing to PD have been investigated in European populations, few studies have investigated these factors in African populations (Walker et al., 2023). Alterations in abundance and composition of the gut microbiome have previously been associated with PD in other populations (Ryman et al., 2023). Given the significant variability observed in gut microbiome findings due to population-level and individual-level genetic and environmental factors (Ryman et al., 2023), investigating the microbiome in understudied populations could be beneficial both for understanding disease pathogenesis and developing tailored intervention strategies for PD.

Materials and methods

Study design and population

Participants were recruited as part of the 'Understanding the SHARED ROOTS of Neuropsychiatric Disorders and Modifiable Risk Factors for Cardiovascular Disease' or SHARED ROOTS (SR; HREC no. N13/08/115; Department of Psychiatry, Faculty of Medicine and Health Sciences, Stellenbosch University, Cape Town, South Africa). Participants with PD were recruited through an existing database of patients at the Movement Disorders Clinic at Tygerberg Academic Hospital, through flyers placed at and referrals from other local healthcare facilities, and print, radio, and web advertisements. Participants with PD were included if they had a UK Brain Bank clinical diagnosis (Hughes et al., 1992) of idiopathic PD made by a neurologist and were 18 years or older. Participants were excluded if they had a history of exposure to neuroleptics, a significant head injury, and atypical forms of PD. Controls (≥ 18 years) were recruited via advertisements (print, radio, and web); active recruitment through a registered nurse visiting various community centres; and word of mouth.

The Mini International Neuropsychiatric Interview (MINI) version 6 (Sheehan et al., 1998) was used to exclude participants with a major psychiatric disorder or who had drug or alcohol use disorders. Additional exclusion criteria for this study included the use of antibiotics within four weeks of stool sampling, severe physical illness, or any other neurological disorder. The WHO STEPwise approach to Surveillance (STEPS)

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instrument (Riley et al., 2016) was used to determine body mass index (BMI), whereas smoking and alcohol habits were assessed with the medical history questionnaire. Participants were classified as current consumers if they smoked or consumed alcohol in the last 6 months.

Sample collection and microbial DNA extraction

Stool sample collection and DNA extraction were done as previously described (Malan-Muller et al., 2022). Stool samples for participants (PD cases: *n* = 16; controls: *n* = 42) were self-collected in pre-analytical sample processing (PSP®) collection tubes (Stratec, Molecular, Birkenfeld, Germany) and stored at -20°C until microbial DNA extraction was performed. Microbial DNA extraction was performed using the PSP Spin Stool DNA Plus kit (Stratec Molecular, Birkenfeld, Germany). The controls added to the microbial DNA extraction included: Biodefense and Emerging Infection Research Resources Repository (BEI resources) mock community A control (BMA, known composition to provide a base to which measurement results can be compared to assess accuracy), community DNA standard (CDS, mixture of genomic DNA extracted from pure cultures) and negative controls. DNA was quantified and quality checked using an ultraviolet-visible (UV-Vis) Spectrometer Nano-Drop 2000 (Thermo Scientific, USA). Amplification of the V4 region of the 16S rRNA gene was conducted using the following primer pair (Caporaso et al., 2012):

515 F (5'-GTGCCAGCMGCCGCGGTAA-3')

806 R (5'-GGACTACHVGGGTWTCTAAT-3')

Library preparation and sequencing was done at Macrogen Inc., (Seoul, Korea) using the Illumina MiSeq® platform (San Diego, CA, USA). The Herculase II Fusion DNA Polymerase kit (Agilent Technologies, Santa Clara, CA, USA) was used to measure the 16S rRNA gene library concentration. The 16S rRNA gene library was then sequenced with 300-bp paired-end reads on an Illumina MiS-eq® sequencing system using a Nextera XT Index Kit v2 (600 cycles; Cat. No. TG-31–1096, Illumina Inc., San Diego, CA, USA). The sample depth was about 60 000 reads per sample and the FASTQ files (forward, reverse, and index read) were generated using the BCL-to-FASTQ file converter bcl2fastq (version 2.17.1.14, Il-lumina, Inc.).

Data preparation and taxa classification

Data preparation, including quality control and taxa classification, was done as previously described (Malan-Muller et al., 2022), using the DADA2 pipeline (https://github.com/benjjneb/dada2) in R Studio (R Core Team, 2020). First, the read quality profiles were inspected, whereafter sequences were filtered and trimmed while maintaining overlap between the forward and reverse sequences to allow appropriate merging at a later stage. Next, a parametric error model was used to estimate error rates and inference of sample composition, after which the sample inference algorithm was applied to the data to determine the number of unique sequences and reduce redundancy. Reads were merged with at least 12 bases in the

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overlap region. After trimming and filtering, an amplicon sequence variant (ASV) table was generated, and chimaeras were identified and removed. The DADA2 algorithm implemented a naïve Bayesian classifier method to taxonomically classify the sequence variants in the ASV table by comparing it to a training set of classified sequences (Callahan *et al.*, 2016). The Ribosomal Database Project (RDP) (Balvočiute & Huson, 2017) was used for the taxonomic classification. The ASV table consisted of the following: 1011870 reads, 3096 taxa, and sparsity of 93.7%. Prior to analyses, we eliminated taxa that were observed in less than 15% of participants from the ASV table as per a previous study (Malan-Muller et al., 2022).

Statistical analysis

To assess the difference between 16 PD cases and 42 controls for categorical variables (sex, current alcohol consumption and current smoker), the Chi-square (χ2) test or Fisher's Exact test (n < 5 for any of the groups) was used. Normality was assessed using the Shapiro-Wilk test (normality: $p > 0.05$). For continuous data (age and BMI) that was not normally distributed, differences between cases and controls were evaluated with the Mann-Whitney U-test (U test) with the results expressed in the form of the median and interquartile range (IQR). The statistically significant threshold was less than 0.05.

Diversity data analysis

Analysis of alpha- and beta-diversity were done with the QIIME2 *q2-diversity* plugin (Bolyen et al., 2019) and differential taxa abundance testing was completed with Analysis of Compositions of Microbiomes with Bias Correction (ANCOM-BC) (Lin & Peddada, 2020) using the *q2-composition* plugin. Statistical significance was assessed using either a p-value or q-value, where the q-value represents significance after correction for multiple testing (Noble, 2009) using the Benjamini-Hochberg method (Haynes, 2013). The significance threshold was set at $\alpha = 0.05$ for all tests (both p-value or q-value).

Alpha-diversity analysis

The microbial richness and evenness within individual samples (alpha-diversity) were assessed using Shannon and Simpson diversity measures (Hurlbert, 1971), as Shannon is more sensitive to species richness and Simpson is more sensitive to the evenness of species (Johnson & Burnet, 2016). Kruskal-Wallis tests were performed to assess the association between alpha-diversity measures and case-control status ($q <$ 0.05).

Covariate selection for beta-diversity

The association between variables including sex, age, BMI, current alcohol consumption, and current smoker, and microbial community abundance (microbiome per-variable-feature association testing) was determined using the Multivariate Association with Linear Models 2 (MaAsLin2) package (Wallen et al., 2022) in R studio (R Core Team, 2020). The covariates were assessed as a collective through the multivariate

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MaAslin2 function using the case-control status "control" as a reference. If the covariate had a statistically significant result for MaAsLin2 ($q < 0.05$), it was included in the subsequent analysis of beta-diversity.

Permutational multivariate analysis of variance (PERMANOVA)

The permutational multivariate analysis of variance (PERMANOVA) was used to evaluate the differences in beta-diversity using Euclidean distance (Anderson, 2001). A PERMANOVA adonis permutation-based statistical test (adonis) test (Oksanen et al., 2018) was performed to assess the contribution of case-control status and covariates on beta-diversity (permutations = 999; $\alpha = 0.05$). Pairwise comparisons were conducted to determine specific associations.

Differential abundance analysis

The differential abundance of taxa associated with case-control status was assessed using ANCOM-BC (Lin & Peddada, 2020). Specific microbial taxa exhibiting significant differential abundance ($q < 0.05$) were reported. Taxa that were enriched or depleted between cases and controls were identified based on the log-fold change provided by the ANCOM-BC diversity analysis (Lin & Peddada, 2020).

Results

Cohort description

Our sample consisted of 16 PD cases (27.6%) and 42 controls (72.4%) (Table 3.1). Age (case: 63.5 (7.6); control: 57.6 (9.0); $p < 0.001$) and BMI (case: 28.4 (11.2); control: 28.5 (9.6); $p = 0.010$) were significantly different between PD cases and controls. Ten (62.5%) of the PD cases and four (9.5%) of the controls also indicated the use of other psychiatric medications for depression and/or sleep.

Table 1: Clinical and demographic characteristics of cohort

Variables	Case (n = 16)	Control (n = 42)	p-value
Sex (female, %)	6 (37.5)	15 (35.7)	1.00 [c]
Age (years)(median, IQR)	63.5 (7.6)	57.6 (9.0)	<0.001 [*U]
BMI (median, IQR)	28.4 (11.2)	28.5 (9.6)	0.010 [*U]
Current alcohol consumption (yes, %)	4 (25.0)	18 (42.9)	0.243 [f]
Current smoker (yes, %)	3 (18.8)	11 (68.8)	0.736 [f]

BMI: body mass index; N: number; IQR: interquartile range (Q3 – Q1)

*[c] Chi-square; [f] Fisher's exact test; [U]: Mann-Whitney U test; * p < 0.05*

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Alpha-diversity measures

Age and BMI had a statistically significant difference between cases and controls and were included in the alpha-diversity analysis. No association was observed for alpha-diversity and case-control status (Shannon $q = 0.414$; Simpson $q = 0.590$), age (Shannon $q = 0.354$; Simpson $q = 0.322$) or BMI (Shannon $q = 0.739$; Simpson $q = 0.871$).

Beta-diversity analysis

Covariates for beta-diversity

Before the beta-diversity analysis, the association between variables (sex, age, BMI, current alcohol consumption, and current smoker) and microbial community abundance was determined using MaAsLin2. However, none of the variables were associated with microbial community abundance. Regardless, sex was included as a covariate for beta-diversity due to the male-to-female incidence ratio of 3:1 commonly reported in Parkinson's disease (Romano et al., 2021). In our cohort, we had 6 female PD cases versus 10 male PD cases and 15 female controls versus 27 male controls.

PERMANOVA and adonis analysis of beta-diversity

We found that beta-diversity was associated with case-control status ($p = 0.008$) as shown in Table 2. Specific taxa differences between cases and controls were subsequently determined with ANCOM-BC.

Table 2: Results of adonis test

Model	R^2	P-value
Status	0.024	0.008*
Sex	0.014	0.205
Residuals	0.957	NA
Total	1.000	NA

** $p < 0.05$*

Differential abundance of taxa associated with Parkinson's disease

The relative abundance differences between PD cases and controls are shown in Figure 1. On the family level, the differential abundance of Lachnospiraceae ($q = 0.028$), Pasteurellaceae ($q = 0.004$), Peptostreptococcaceae ($q < 0.001$), and Clostridiaceae ($q = 0.015$) were all depleted in PD cases compared to controls. PD was associated with the decreased differential abundance of *Faecalibacterium* ($q = 0.007$), *Roseburia* ($q = 0.030$), *Dorea* ($q = 0.015$), *Clostridium* ($q = 0.016$) and *Veillonella* ($q < 0.001$). Moreover, *Victivallis* ($q < 0.001$) and *Akkermansia* ($q = 0.015$) were enriched in PD cases versus controls.

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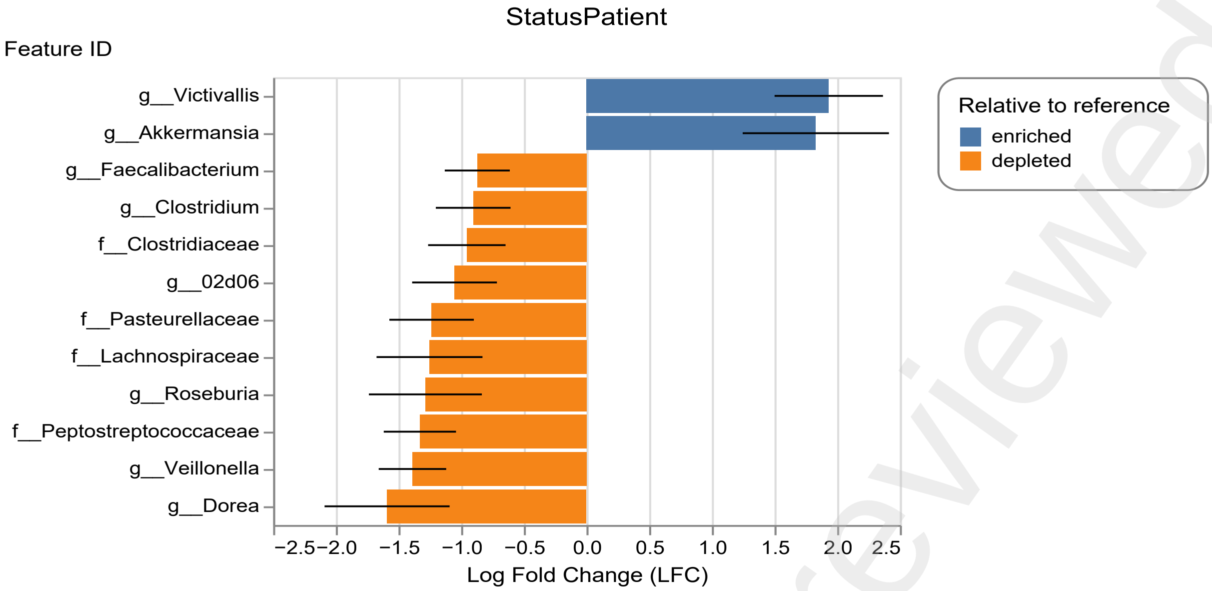

Figure 1: Statistically significant relative abundance differences between PD cases and controls. The relative abundance of taxa that were statistically different in PD cases compared to controls was assessed using ANCOM-BC. Enriched or depleted relative abundances between cases and controls were identified based on the log-fold change and considered statistically significant at a threshold of 0.05 after correction for multiple testing based on the Benjamini-Hochberg method. On family level, the relative abundance of *Lachnospiraceae* ($q = 0.028$), *Pasteurellaceae* ($q = 0.004$), *Peptostreptococcaceae* ($q < 0.001$), and *Clostridiaceae* ($q = 0.015$) were depleted in PD cases. Furthermore, the decreased relative abundance of *Faecalibacterium* ($q = 0.007$), *Roseburia* ($q = 0.030$), *Dorea* ($q = 0.015$), *Clostridium* ($q = 0.016$) and *Veillonella* ($q < 0.001$) was associated with PD at the genus level. The relative abundance of *Victivallis* ($q < 0.001$) and *Akkermansia* ($q = 0.015$) were enriched in PD cases compared to the controls. *g__: genus; f__: family*

Discussion

We investigated the association between the gut microbiome and PD in a South African sample as the identification of microbiome markers may lead to novel avenues for treatment (Walker et al., 2023). We did not find a statistically significant association between alpha-diversity and PD status. We did, however, find a statistically significant association between PD status and beta-diversity.

Of the microbial genera differentially abundant in PD cases versus controls in our study, previous studies corroborated the differences in relative abundance for *Akkermansia* (Hirayama & Ohno, 2021; Ryman et al., 2023), *Faecalibacterium,* and *Roseburia* (Hirayama & Ohno, 2021). Moreover, Nishiwaki and colleagues observed an increase in the relative abundance of *Akkermansia* and a decrease in *Faecalibacterium* and *Roseburia* after conducting a meta-analysis on 16S rRNA samples from 223 PD cases and 137 controls (Nishiwaki et al., 2020), in line with our findings. *Akkermansia* has mucin layer-degrading and LPS-producing properties (Lei et al., 2021) which could contribute to mucin degradation and increase in intestinal permeability (Hirayama & Ohno, 2021). Increased intestinal permeability may allow bacterial translocation from the gut into the blood, which has been found to trigger the release of pro-inflammatory cytokines

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(Nishiwaki et al., 2020; Hirayama & Ohno, 2021; Lei et al., 2021; Ryman et al., 2023). Therefore, it is suggested that an increased relative abundance of *Akkermansia* in the gut could contribute to a pro-inflammatory environment (Lei et al., 2021; Takahashi et al., 2022). Both *Faecalibacterium* and *Roseburia* are butyrate-producing genera with anti-inflammatory properties and a decrease in the relative abundance of these taxa could contribute to the overexpression of pro-inflammatory cytokines (Nishiwaki et al., 2020) and a lower concentration of SCFA being available in the gut to support gut barrier integrity (Nishiwaki et al., 2020; Hirayama & Ohno, 2021; Romano et al., 2021; Li et al., 2023).

It is noteworthy that the bacterial metabolites, SCFA and LPS produced by the abovementioned taxa, have been associated with increased intestinal permeability and inflammation in PD (Hirayama & Ohno, 2021; Romano et al., 2021). The inflammatory response linked to alterations in SCFA and LPS levels (Aho et al., 2021; Hirayama & Ohno, 2021; Romano et al., 2021) may also trigger microglial activation, a key driver in PD progression (Romano et al., 2021). Butyrate and other SCFAs are essential for normal microglia development (Aho et al., 2021; Romano et al., 2021), suggesting that alteration in SCFA levels may contribute to PD development. Furthermore, LPS-induced elevation in pro-inflammatory cytokines could disrupt neuronal activity in limbic brain regions, potentially amplifying CNS cytokine production (Diaz et al., 2022). Increased levels of LPS, LPS-induced elevation in pro-inflammatory cytokines, and decreased levels of SCFA in the blood could compromise the integrity of the BBB (Ryman et al., 2023). A compromised BBB could facilitate the translocation of pathogens and pro-inflammatory agents into the brain environment, thereby inducing inflammation in the brain (Ryman et al., 2023). Moreover, LPS, together with a compromised BBB, have been found to be associated with increased movement of α-synuclein aggregates that further contributes to microglial activation and neurotoxicity in the brain (Nishiwaki et al., 2020; Hirayama & Ohno, 2021).

Although an altered gut microbiome in patients with PD compared to controls was observed in our study, and previous studies (Takahashi et al., 2022; Li et al., 2023), no single microbial profile has been reported to be associated with the disorder. This could be attributed to considerable variability in gut microbial findings associated with disease observed between studies possibly due to sampling protocols, country of origin, sample storage, DNA extractions, and sequencing strategies (Ryman et al., 2023). However, after accounting for these possible inconsistencies mentioned, shared factors between studies include a decline in the relative abundance of butyrate-producing taxa (*Roseburia*, *Faecalibacterium*, Lachnospiraceae)(Nishiwaki et al., 2020; Hirayama & Ohno, 2021; Lei et al., 2021; Romano et al., 2021; Takahashi et al., 2022) and an increase in the relative abundance of taxa relating to mucin layer-degrading and LPS producing taxa, such as *Akkermansia* (Aho et al., 2021; Takahashi et al., 2022; Ryman et al., 2023).

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The lack of case-control or cohort studies until recently has hindered comprehensive understanding and appropriate management of PD in South Africa. Studies conducted in South Africa have highlighted deficiencies in both information and treatment options for PD, leading to potential misdiagnosis and inadequate care. Moreover, the scarcity of medical specialists in Africa, limited training facilities and skilled professionals often emigrating to developed countries, have exacerbated the situation (Walker et al., 2023).

Limitations

While these results suggest associations between specific microbial taxa and patients with PD in our sample, there are methodological limitations. The small sample size and low power (0.387) could have contributed to Type II error (false negative results). PD patients may have additional barriers to providing stool samples, such as constipation and mobility challenges, contributing to sample size. Second, the patient sample was recruited from a localised region of the country and may not be representative of all PD patients in SA. Larger studies that are more geographically representative of the SA population are needed to validate and strengthen these findings.

Conclusion

This exploratory study contributes to growing evidence of the role that the gut microbiome may play in PD. In line with previous studies, our study found PD to be associated with a reduction in butyrate-producing bacteria (*Faecalibacterium* and *Roseburia*) and an increase in mucin-degrading bacteria (*Akkermansia*). The differences in these taxa abundance might be associated with heightened gut intestinal permeability and low-grade systemic inflammation due to increased levels of LPS and decreased levels of SCFA. However, longitudinal studies are needed to address the question of whether these microbiome differences are a risk factor for or are consequent to PD. Future research should longitudinally investigate the emergence of non-motor symptoms, alterations in gut microbiome composition over time, and the contributions of other role players of the MGB axis, such as alterations to SCFA, pro-inflammatory cytokines or neurotransmitters, the tryptophan metabolism, and the hypothalamic-pituitary-adrenal (HPA) axis, in patients with subclinical symptoms, to name a few possibilities. The discovery of specific microbiome biomarkers could advance our understanding of the gut-brain axis, PD pathogenesis and the invisible string between the gut microbiome and disease.

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Authors' contributions

CR: Project administration, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft

SB: Writing – Review & Editing

LLvdH: Data Curation, Writing – Review & Editing

JC: Writing – Review & Editing

SS: Writing – Review & Editing, Funding acquisition

RP: Conceptualization, Supervision, Writing – Review & Editing

SMJH: Conceptualization, Project administration, Funding acquisition, Supervision, Writing – Review & Editing

Abbreviations

ASV: amplicon sequence variant

α-synuclein: alpha-synuclein

ANCOM-BC: Analysis of Compositions of Microbiomes with Bias Correction

BBB: blood-brain barrier

BMI: body mass index

CNS: central nervous system

ENS: enteric nervous system

LPS: lipopolysaccharides

MaAsLin2: Multivariate Association with Linear Models 2

MGB: microbiome-gut-brain

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MINI: Mini International Neuropsychiatric Interview

PCoA: principal coordinates analysis

PD: Parkinson's disease

PERMANOVA: permutational multivariate analysis of variance

PSP: pre-analytical sample processing

RDP: Ribosomal Database Project

SCFA: short-chain fatty acid

SR: Understanding the SHARED ROOTS of Neuropsychiatric Disorders and Modifiable Risk Factors for Cardiovascular Disease' or SHARED ROOTS

STEPS: WHO STEPwise approach to Surveillance

$\chi 2$: Chi-square

Declaration of Competing Interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Ethical approval

This study is a sub-study of the project entitled "Understanding the SHARED ROOTS of Neuropsychiatric Disorders and modifiable risk factors of cardiovascular disease" (ethics number: S22/05/094 (PhD) substudy N13/08/115) which was approved by the Stellenbosch University Health Research Ethics Committee 1 (HREC1). The patients/participants provided their written informed consent to participate in this study

Availability of data and materials

The datasets presented in this article are not readily available due to ethical and legal restrictions. Requests to access the datasets should be directed to SMJH (smjh@sun.ac.za). The authors are open to collaborating and sharing data within the limits of ethical review restrictions and data transfer policies of Stellenbosch University.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the first author(s) used ChatGPT in order to improve readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Tables

[TABLE 1]

Table 1: Clinical and demographic characteristics of cohort

Variables	Case (n = 16)	Control (n = 42)	p-value
Sex (female, %)	6 (37.5)	15 (35.7)	1.00 [c]
Age (years)(median, IQR)	63.5 (7.6)	57.6 (9.0)	<0.001 *[mw]
BMI (median, IQR)	28.4 (11.2)	28.5 (9.6)	0.010 *[mw]
Current alcohol consumption (yes, %)	4 (25.0)	18 (42.9)	0.243 [f]
Current smoker (yes, %)	3 (18.8)	11 (68.8)	0.736 [f]

BMI: body mass index; N: number; IQR: interquartile range (Q3 – Q1)
*[c] Chi-square; [f] Fisher's exact test; [mw] Mann-Whitney U test; * $p < 0.05$*

[TABLE 2]

Table 2: Assessment of contribution of metadata to the beta-diversity of PD cases and controls

Model	R^2	P-value
Status	0.024	0.008*
Sex	0.014	0.205
Residuals	0.957	NA
Total	1.000	NA

* $p < 0.05$

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